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THIN RESISTOR AND MANUFACTURING METHOD THEREOF

Abstract

A thin resistor includes an insulating layer, a resistive layer, a pair of internal electrodes, a protective layer, and a pair of external electrodes. The resistive layer is disposed on the insulating layer, in which the resistive layer includes a pair of recesses, and the pair of recesses is located on two opposite ends of the resistive layer, respectively. The pair of internal electrodes is respectively disposed in the pair of recesses and on the resistive layer, and top surfaces of the pair of internal electrodes are higher than a top surface of the resistive layer. The protective layer covers a portion of the resistive layer and portions of the internal electrodes. The pair of external electrodes is electrically connected to the pair of internal electrodes.

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Background/Summary

RELATED APPLICATION

[0001] This application claims priority to Taiwan Application Serial Number 113106514, filed Feb. 23, 2024, which is herein incorporated by reference.

BACKGROUND

Field of Invention

[0002] The present disclosure relates to a thin resistor and a manufacturing method thereof. More particularly, the present disclosure relates to the thin resistor with a pair of internal electrodes embedded in a resistive layer.

Description of Related Art

[0003] In the field of current sense resistors, a resistance value of a resistor can usually be reduced by increasing a thickness of a resistive layer or reducing the spacing between a pair of internal electrodes. However, a thicker resistive layer can increase an overall size of the resistor, which is disadvantageous to subsequent applications of the resistor. In addition, a narrower spacing between the pair of internal electrodes is disadvantageous to fine adjustment of the resistance value of the resistor, and is also disadvantageous to the control of a temperature coefficient of resistance (TCR) of the resistor.

[0004] Furthermore, if the TCR of the resistor is reduced, it will have to increase a thickness of the pair of internal electrodes, so as to reduce the contribution of the pair of internal electrodes to the resistance value of the overall resistor. However, the greater thickness of the pair of inner electrodes will increase the overall size of the resistor, which is disadvantageous to subsequent applications of the resistor.

[0005] For the foregoing reason, there is a need to solve the above-mentioned problems by providing a resistor and a manufacturing method thereof.

SUMMARY

[0006] The thin resistor of the present disclosure has a pair of internal electrodes partially embedded in a resistive layer. Therefore, an overall thickness of the thin resistor can be reduced, which is beneficial to the subsequent applications of the thin resistor, as well as the better control of TCR of the thin resistor. Furthermore, because the internal electrodes of the present disclosure are partially embedded in the resistive layer, the current density at both ends of the pair of internal electrodes of the thin resistor of the present disclosure is less (the current between the resistive layer and the internal electrodes is relatively smooth) than the one of the pair of the conventional internal electrodes that does not embed in the resistive layer, resulting in reduced heat accumulation and increased heat conduction capability.

[0007] In at least one embodiment of the present disclosure, a thin resistor is provided. The thin resistor includes an insulating layer, a resistive layer, a pair of internal electrodes, a protective layer, and a pair of external electrodes. The resistive layer is disposed on the insulating layer, in which the resistive layer includes a pair of recesses respectively located on two opposite ends of the resistive layer. The pair of internal electrodes is respectively disposed in the pair of recesses and on the resistive layer, and top surfaces of the pair of internal electrodes are higher than a top surface of the resistive layer. The protective layer covers a portion of the resistive layer and portions of the pair of internal electrodes. The pair of external electrodes electrically connects the pair of internal electrodes.

[0008] In at least one embodiment of the present disclosure, the resistive layer has a resistance long

side and a resistance short side, the resistance long side has a resistance length L, the resistance short side has a resistance width W, each of the pair of recesses has a recess long side and a recess short side, and a shortest vertical distance between the resistance long side and the recess short side is in a range from 0 μm to $\frac{1}{6} W$.

[0009] In at least one embodiment of the present disclosure, the resistance length L is in a range from 0.5 mm to 6.5 mm.

[0010] In at least one embodiment of the present disclosure, the resistance width W is in a range from 0.25 mm to 3.25 mm.

[0011] In at least one embodiment of the present disclosure, a shortest vertical distance between the resistance short side and the recess long side is in a range from 0 μm to $\frac{1}{10} L$.

[0012] In at least one embodiment of the present disclosure, the recess short side has a recess width, and the recess width is in a range from $\frac{1}{7} L$ to $\frac{1}{3} L$.

[0013] In at least one embodiment of the present disclosure, a shortest vertical distance between two recesses is in a range from $\frac{1}{25} L$ to $\frac{1}{3} L$.

[0014] In at least one embodiment of the present disclosure, the resistive layer has a resistance thickness T, each of the pair of recesses has a recess depth, and the recess depth is in a range from $\frac{1}{10} T$ to $\frac{2}{3} T$.

[0015] In at least one embodiment of the present disclosure, the resistance thickness T is in a range from 0.075 mm to 0.3 mm.

[0016] In at least one embodiment of the present disclosure, top surfaces of the pair of internal electrodes are at least 10 μm higher than a top surface of the resistive layer.

[0017] In at least one embodiment of the present disclosure, the pair of external electrodes includes a pair of front electrodes, and top surfaces of the front electrodes are at least 5 μm higher than a top surface of the protective layer.

[0018] In at least one embodiment of the present disclosure, the pair of external electrodes further includes a pair of nickel layers and a pair of tin layers. The pair of nickel layers covers the pair of front electrodes. The pair of tin layers covers the pair of nickel layers.

[0019] In at least one embodiment of the present disclosure, a manufacturing method of a thin resistor is provided. A resistive layer is formed on an insulating layer. Portions of the insulating layer are recessed to form a pair of recesses, in which the pair of recesses is respectively located on opposite sides of the resistive layer. A pair of internal electrodes is formed in the pair of recesses, in which top surfaces of the pair of internal electrodes are higher than a top surface of the resistive layer. A protective layer is formed to cover a first portion of the resistive layer and portions of the pair of internal electrodes. A pair of external electrodes is formed to electrically connect the pair of internal electrodes.

[0020] In at least one embodiment of the present disclosure, after forming the pair of internal electrodes, performing a trimming operation on the resistive layer.

[0021] In at least one embodiment of the present disclosure, the pair of internal electrodes is formed by electroplating.

[0022] In at least one embodiment of the present disclosure, the pair of recesses is formed by an etching process.

[0023] In at least one embodiment of the present disclosure, an etching solution used in the etching process includes copper chloride, sulfuric acid, phosphoric acid, nitric acid, or combinations thereof.

[0024] In at least one embodiment of the present disclosure, further comprising: before recessing the portions of the insulating layer to form the pair of recesses, forming a mask layer on the resistive layer; and after recessing the portions of the insulating layer to form the pair of recesses, removing the mask layer to expose the top surface of the resistive layer.

[0025] In at least one embodiment of the present disclosure, further comprising: before forming the pair of internal electrodes in the pair of recesses, forming a mask layer to cover a second portion of

the resistive layer and expose the pair of recesses; and after forming the pair of internal electrodes in the pair of recesses, removing the mask layer to expose the top surface of the resistive layer.

[0026] In at least one embodiment of the present disclosure, further comprising: after removing the mask layer to expose the top surface of the resistive layer, performing a trimming operation on the resistive layer to form a trimming groove.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0028] FIG. 1 and FIG. 2 are cross-sectional views of a thin resistor according to one embodiment of the present disclosure.

[0029] FIG. 3A, FIG. 4A, FIG. 5A, FIG. 6A, FIG. 7A, and FIG. 8A are three-dimensional diagrams of a thin resistor at various stages in accordance with one embodiment of the present disclosure.

[0030] FIG. 3B, FIG. 4B, FIG. 5B, FIG. 6B, FIG. 7B, and FIG. 8B are cross-sectional views obtained from lines A-A' of FIG. 3A, FIG. 4A, FIG. 5A, FIG. 6A, FIG. 7A, and FIG. 8A, respectively.

[0031] FIG. 4C is a three-dimensional diagram of a resistive layer of FIG. 4A.

DETAILED DESCRIPTION

[0032] The following disclosure provides many different embodiments, or examples, for implementing different features of the present disclosure. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact.

[0033] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0034] In the present specification, a range represented by “one value to another value” is a summary representation that avoids enumerating all the values in the range in the specification. Therefore, the recitation of a particular numerical range covers any numerical value within the numerical range and the smaller numerical range defined by any numerical values within the numerical range, as if the arbitrary value and the smaller numerical range are expressly stated in the specification.

[0035] FIG. 1 and FIG. 2 are cross-sectional views of a thin resistor **100** according to one embodiment of the present disclosure. Referring to FIG. 1, the thin resistor **100** includes an insulating layer **110**, a resistive layer **120**, and a pair of internal electrodes **130**. The resistive layer **120** is disposed on the insulating layer **110**, in which the resistive layer **120** includes a trimming groove **122**. The resistive layer **120** includes a pair of recesses **R**, and the pair of recesses **R** is respectively located on two opposite ends of the resistive layer **120**. The pair of internal electrodes

130 is respectively disposed in the pair of recesses **R** and on the resistive layer **120**. In other words, portions of the internal electrodes **130** are embedded in the resistive layer **120**, and other portions of the internal electrodes **130** protrude from a top surface **120s** of the resistive layer **120**.

[0036] In the embodiment of FIG. 1, top surfaces **130s** of the internal electrodes **130** are higher than the top surface **120s** of the resistive layer **120**. In some embodiments, a gap **G1** between the top surfaces **130s** of the internal electrodes **130** and the top surface **120s** of the resistive layer **120** is at least 10 μm . If the gap **G1** was less than 10 μm , it would be disadvantageous to adjust a resistance value of the internal electrodes **130** by using a probe in the subsequent process. A bottom surface of the pair of internal electrodes **130** is lower than the top surface **120s** of the resistive layer **120**. In some embodiments, an overall thickness of the internal electrodes **130** is at least 100 μm . It should be noted that the “overall thickness” of the internal electrodes **130** herein includes the sum of a thickness of the internal electrodes **130** in the recesses **R** and a thickness of the internal electrodes **130** higher than the top surface **120s** of the resistive layer **120**.

[0037] As shown in FIG. 1, the thin resistor **100** further includes a protective layer **140** and a pair of external electrodes **150** (may be referred to as terminal electrodes). The protective layer **140** covers a portion of the resistive layer **120** and portions of the internal electrodes **130**. The pair of external electrodes **150** electrically connects the pair of internal electrodes **130**, in which the pair of external electrodes **150** includes a pair of front electrodes **152**, a pair of nickel layers **154**, and a pair of tin (Sn) layers **156**. The pair of front electrodes **152** covers portions of the internal electrodes **130** and two side surfaces of the resistive layer **120**. The pair of nickel layers **154** covers the pair of front electrodes **152**, and the pair of tin layers **156** covers the pair of nickel layers **154**.

[0038] In the embodiment of FIG. 1, the top surfaces **152s** of the pair of front electrodes **152** are higher than a top surface **140s** of the protective layer **140**. In some embodiments, a gap **G2** between the top surfaces **152s** of the pair of front electrodes **152** and the top surface **140s** of the protective layer **140** is at least 5 μm . If the gap **G2** is less than 5 μm , it would be disadvantageous to soldering of the thin resistor **100** in the subsequent process.

[0039] Referring to FIG. 2, FIG. 2 is an inverted structure of FIG. 1, and the thin resistor **100** in FIG. 2 is disposed on pads **210**, in which the dashed arrows represent the direction of current. It can be known from FIG. 2 that because the internal electrodes **130** of the disclosed thin resistor **100** are partially embedded in the resistive layer **120**, the current between the resistive layer **120** and internal electrodes **130** is relatively smooth compared with the internal electrodes that do not embed in the resistive layer. Therefore, heat accumulation at both ends of the internal electrodes **130** can be reduced, thereby increasing the heat conduction capability. Furthermore, because the internal electrodes **130** are partially embedded in the resistive layer **120**, the overall thickness of the resistor can be reduced, which is beneficial to the subsequent applications of the resistor and is beneficial for controlling TCR of the resistor.

[0040] FIG. 3A, FIG. 4A, FIG. 5A, FIG. 6A, FIG. 7A, and FIG. 8A are three-dimensional diagrams of the thin resistor **100** at various stages in accordance with one embodiment of the present disclosure. FIG. 3B, FIG. 4B, FIG. 5B, FIG. 6B, FIG. 7B, and FIG. 8B are cross-sectional views obtained from lines A-A' of FIG. 3A, FIG. 4A, FIG. 5A, FIG. 6A, FIG. 7A, and FIG. 8A, respectively.

[0041] Referring to FIG. 3A and FIG. 3B, the resistive layer **120** is formed on the insulating layer **110**. In some embodiments, the insulating layer **110** is a carrier with a single-sided adhesive. In some embodiments, the carrier includes an insulating material, such as polyimide (PI), glass fiber epoxy resin (FR4), ceramic substrate, or glass substrate, but is not limited thereto. In other embodiments, the insulating layer **110** (carrier) and the resistive layer **120** are bonded together using a double-sided adhesion layer.

[0042] In some embodiments, the resistive layer **120** includes a metallic alloy material, such as a MnCu alloy, a CuNi alloy, a CuMnNi alloy, a CuMnSn alloy, a NiCrAl alloy, a NiCrAlSi alloy, or a FeCrAl alloy, but is not limited thereto.

[0043] Still referring to FIG. 3A and FIG. 3B, the mask layer **310** is formed on the resistive layer **120**, so as to cover a portion of the resistive layer **120** and expose other portions of the top surfaces **120s** of the resistive layer **120**. In some embodiments, the patterned mask layer **310** is formed by printing, laminating, coating, and/or photolithography. A material of the mask layer **310** could include a photoresist material, a removable adhesive film, or ink. It could be understood that the exposed top surfaces **120s** are areas where subsequent etching operations could be performed. Although the mask layer **310** shown in FIG. 3A includes two rectangles, the pattern of the mask layer **310** could be adjusted according to actual needs.

[0044] Referring to FIG. 4A and FIG. 4B, portions of the insulating layer **110** are recessed to form the pair of recesses **R**, in which the pair of recesses **R** is respectively located on two opposite ends of the resistive layer **120**. Specifically, the pair of recesses **R** is formed by an etching process. In some embodiments, an etching solution used in the etching process includes copper chloride, sulfuric acid, phosphoric acid, nitric acid, or combinations thereof. It could be understood that two recesses **R** of the present disclosure are symmetrically arranged and have the same or similar dimensions.

[0045] Referring to FIG. 4C, FIG. 4C is a three-dimensional diagram of the resistive layer **120** of FIG. 4A. The resistive layer **120** has a resistance long side **A** and a resistance short side **B**, in which the resistance long side **A** has a resistance length **L**, and the resistance short side **B** has a resistance width **W**. Each of the recesses **R** has a recess long side **a** and a recess short side **b**. In some embodiments, the resistance length **L** is in a range from 0.5 mm to 6.5 mm. In some embodiments, the resistance width **W** is in a range from 0.25 mm to 3.25 mm.

[0046] In some embodiments, a shortest vertical distance **D1** between the resistance long side **A** and the recess short side **b** is in a range from 0 μm to $\frac{1}{6} W$. If the distance **D1** was greater than $\frac{1}{6} W$, it would be disadvantageous to finely adjust the resistance value of the resistive layer **120**.

[0047] In some embodiments, a shortest vertical distance **D2** between the resistance short side **B** and the recess long side **a** is in a range from 0 μm to $\frac{1}{10} L$. If the distance **D2** was greater than $\frac{1}{10} L$, it would be disadvantageous to finely adjust the resistance value of the resistive layer **120**. It could be understood that a cross-section of the resistive layer **120** has a T-like profile when both the distance **D1** and the distance **D2** are 0 μm .

[0048] In some embodiments, a shortest vertical distance **D3** between two recesses **R** is in a range from $\frac{1}{25} L$ to $\frac{1}{3} L$.

[0049] Still referring to FIG. 4C, in some embodiments, the recess short side **b** has a recess width **Rw**, and a recess width **Rw** is in a range from $\frac{1}{7} L$ to $\frac{1}{3} L$. The wider the recess width **Rw**, the bigger the internal electrodes (i.e., the internal electrodes **130** in FIG. 1) formed in the subsequent processes and the smaller the overall resistance value of the resistor. The narrower the recess width **Rw**, the smaller the internal electrodes (i.e., the internal electrodes **130** in FIG. 1) formed in the subsequent processes and the greater the overall resistance value of the resistor.

[0050] Still referring to FIG. 4C, the resistive layer **120** has a resistance thickness **T**. In some embodiments, the resistance thickness **T** is in a range from 0.075 mm to 0.3 mm. In some embodiments, a recess depth of the recess **R** is in a range from $\frac{1}{10} T$ to $\frac{2}{3} T$. If the recess depth is in a range from $\frac{1}{10} T$ to $\frac{2}{3} T$, the current flowing through the resistive layer **120** and the internal electrodes **130** (refer to FIG. 2) is smooth, which can reduce heat accumulation and increase the heat conduction capability and the heat dissipation capability. If the recess depth was less than $\frac{1}{10} T$, the thin resistor of the present disclosure could not be formed, and the current flowing through the resistive layer **120** and the internal electrodes **130** would excessively concentrate at the ends of the internal electrodes **130**, which is disadvantageous to heat conduction. If the recess depth was greater than $\frac{2}{3} T$, the support of the resistive layer **120** might be insufficient, thereby causing the bended resistor.

[0051] After the pair of recesses **R** shown in FIG. 4A and FIG. 4B is formed, the mask layer **310** is removed, so as to expose the top surface **120s** of the resistive layer **120**. The mask layer **310** can be

removed by using film-removing liquid.

[0052] Referring to FIG. 5A and FIG. 5B, the mask layer **510** is formed on the resistive layer **120**, so as to cover a portion of the resistive layer **120** and expose other portions of the top surfaces **120s** of the resistive layer **120**, in which the recesses **R** are also exposed by the mask layer **510**. It is noted that two ends of the top surface **120s** of the resistive layer **120** are also exposed by the mask layer **510**. In some embodiments, the patterned mask layer **510** is formed by printing, laminating, coating, and/or photolithography. A material of the mask layer **510** could include a photoresist material, a removable adhesive film, or ink.

[0053] Referring to FIG. 6A and FIG. 6B, the pair of internal electrodes **130** is formed in the pair of recesses **R**, in which the top surfaces **130s** of the internal electrodes **130** are higher than the top surface **120s** of the resistive layer **120**. The pair of internal electrodes **130** is formed by electroplating. A material of the internal electrodes **130** includes copper. It could be understood that the pair of internal electrodes **130** is firstly formed by filling the pair of recesses **R** (refer to FIG. 5A and FIG. 5B), and then continuing to form copper material (or other material of internal electrodes), so as to form the pair of internal electrodes **130** protruding from the top surface **120s** of the resistive layer **120**. As shown in FIG. 6B, a cross-section of each of the internal electrodes **130** has a T-like profile.

[0054] After the internal electrodes **130** shown in FIG. 6A and FIG. 6B are formed, the mask layer **510** is removed, so as to expose the top surface **120s** of the resistive layer **120**. The mask layer **510** can be removed by using film-removing liquid.

[0055] Referring to FIG. 7A and FIG. 7B, a trimming operation is performed on the resistive layer **120** to form a trimming groove **122**. In some embodiments, trimming groove **122** is formed by laser or physical processing.

[0056] Referring to FIG. 8A and FIG. 8B, the protective layer **140** is formed, so as to cover a portion of the resistive layer **120** and portions of the internal electrodes **130**. In some embodiments, a material of the protective layer **140** includes epoxy or resin. In some embodiments, the insulating protective layer **140** is formed by printing or photolithography.

[0057] After the protective layer **140** is formed, the pair of external electrodes **150** is formed to electrically connect the pair of internal electrodes **130**, as shown in FIG. 1. The nickel layer **154** and the tin layer **156** are formed by electroplating.

[0058] It could be understood that the dimensions (i.e., the resistance length **L**, the resistance width **W**, and the resistance thickness **T**) of the disclosed resistive layer **120** can be arbitrarily enlarged or reduced according to the actual applications of the resistor and the actual needs of the product, and is not particularly limited.

[0059] In summary, the thin resistor of the present disclosure has the pair of internal electrodes partially embedded in the resistive layer. Therefore, the overall thickness of the film resistor can be reduced, which is beneficial to the subsequent applications of the thin resistor, as well as the better control of TCR of the thin resistor. Furthermore, because the internal electrodes of the present disclosure are partially embedded in the resistive layer, the current density at both ends of the pair of internal electrodes of the thin resistor of the present disclosure is less (the current between the resistive layer and the internal electrodes is relatively smooth) than the one of the pair of the conventional internal electrodes that does not embed in the resistive layer, resulting in reduced heat accumulation and increased heat conduction capability.

[0060] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that

they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A thin resistor, comprising: an insulating layer; a resistive layer disposed on the insulating layer, wherein the resistive layer includes a pair of recesses located on two opposite ends of the resistive layer, respectively; a pair of internal electrodes respectively disposed in the pair of recesses and on the resistive layer, wherein top surfaces of the pair of internal electrodes are higher than a top surface of the resistive layer; a protective layer covering a portion of the resistive layer and portions of the pair of internal electrodes; and a pair of external electrodes electrically connecting the pair of internal electrodes.
2. The thin resistor of claim 1, wherein the resistive layer has a resistance long side and a resistance short side, the resistance long side has a resistance length L , the resistance short side has a resistance width W , each of the pair of recesses has a recess long side and a recess short side, and a shortest vertical distance between the resistance long side and the recess short side is in a range from $0\text{ }\mu\text{m}$ to $\frac{1}{6} W$.
3. The thin resistor of claim 2, wherein the resistance length L is in a range from 0.5 mm to 6.5 mm .
4. The thin resistor of claim 2, wherein the resistance width W is in a range from 0.25 mm to 3.25 mm .
5. The thin resistor of claim 2, wherein a shortest vertical distance between the resistance short side and the recess long side is in a range from $0\text{ }\mu\text{m}$ to $\frac{1}{10} L$.
6. The thin resistor of claim 2, wherein the recess short side has a recess width, and the recess width is in a range from $\frac{1}{7} L$ to $\frac{1}{3} L$.
7. The thin resistor of claim 2, wherein a shortest vertical distance between two recesses is in a range from $\frac{1}{25} L$ to $\frac{1}{3} L$.
8. The thin resistor of claim 1, wherein the resistive layer has a resistance thickness T , each of the pair of recesses has a recess depth, and the recess depth is in a range from $\frac{1}{10} T$ to $\frac{2}{3} T$.
9. The thin resistor of claim 8, wherein the resistance thickness T is in a range from 0.075 mm to 0.3 mm .
10. The thin resistor of claim 1, wherein the top surfaces of the pair of internal electrodes are at least $10\text{ }\mu\text{m}$ higher than the top surface of the resistive layer.
11. The thin resistor of claim 1, wherein the pair of external electrodes comprises a pair of front electrodes, and top surfaces of the pair of front electrodes are at least $5\text{ }\mu\text{m}$ higher than a top surface of the protective layer.
12. The thin resistor of claim 11, wherein the pair of external electrodes further comprises: a pair of nickel layers covering the pair of front electrodes; and a pair of tin layers covering the pair of nickel layers.
13. A manufacturing method of a thin resistor, comprising: forming a resistive layer on an insulating layer; recessing portions of the insulating layer to form a pair of recesses, wherein the pair of recesses is located on two opposite ends of the resistive layer, respectively; forming a pair of internal electrodes in the pair of recesses, wherein top surfaces of the pair of internal electrodes are higher than a top surface of the resistive layer; forming a protective layer to cover a first portion of the resistive layer and portions of the pair of internal electrodes; and forming a pair of external electrodes to electrically connect the pair of internal electrodes.
14. The manufacturing method of the thin resistor of claim 13, wherein after forming the pair of internal electrodes, performing a trimming operation on the resistive layer.
15. The manufacturing method of the thin resistor of claim 13, wherein the pair of internal

electrodes is formed by electroplating.

16. The manufacturing method of the thin resistor of claim 13, wherein the pair of recesses is formed by an etching process.

17. The manufacturing method of the thin resistor of claim 16, wherein an etching solution used in the etching process comprises copper chloride, sulfuric acid, phosphoric acid, nitric acid, or combinations thereof.

18. The manufacturing method of the thin resistor of claim 13, further comprising: before recessing the portions of the insulating layer to form the pair of recesses, forming a mask layer on the resistive layer; and after recessing the portions of the insulating layer to form the pair of recesses, removing the mask layer to expose the top surface of the resistive layer.

19. The manufacturing method of the thin resistor of claim 13, further comprising: before forming the pair of internal electrodes in the pair of recesses, forming a mask layer to cover a second portion of the resistive layer and expose the pair of recesses; and after forming the pair of internal electrodes in the pair of recesses, removing the mask layer to expose the top surface of the resistive layer.

20. The manufacturing method of the thin resistor of claim 19, further comprising: after removing the mask layer to expose the top surface of the resistive layer, performing a trimming operation on the resistive layer to form a trimming groove.
